

Experiment No: 02
Experiment Name: To verify Ohm's law in a simple circuit and calculate the value of unknown resistance

Course Code: EEE-1209
Course Title: Electrical Circuit lab

Submitted to:

Ms. Noshin Tahsin

Lecturer

Department of Computer Science and Engineering
Bangabandhu Sheikh Mujibur Rahman University
Kishoreganj, Bangladesh

Mr. Md. Sabab Zulfiker

Lecturer

Department of Computer Science and Engineering
Bangabandhu Sheikh Mujibur Rahman University
Kishoreganj, Bangladesh

Submitted By:

Group-02:

Names	ID No
Md. Sabbir Mia	202122104017
Amir Hamja Seman	202122104018
Md. Reduanur Rahman Rifat	202122104020
Md. Ehasan Mahmud	202122104026



Date of Submission: 26/09/2023

Department of Computer Science and Engineering,
Bangabandhu Sheikh Mujibur Rahman University
Kishoreganj, Bangladesh

Experiment: To verify Ohm's law in a simple circuit and calculate the value of unknown Resistor.

Ohm's law Theory: The fundamental relationship among voltage, current and resistance was discovered by George Simon Ohm, which is known as Ohm's law. Ohm's law states that, if temperature and other physical conditions of a conductor remain unchanged then current flowing through the conductor is directly proportional to the voltage difference between two terminals of the conductor.

mathematically,

$$I \propto V$$

$$I = GV$$

G is a constant called conductance

$$G = \frac{1}{R}$$

R is called resistance, so that

$$R = \frac{V}{I}$$

Required Equipments:

1. Bread Board
2. Resistor
3. Volt Multimeter
4. Variable DC power supply
5. Connecting wires

Circuit Construction:

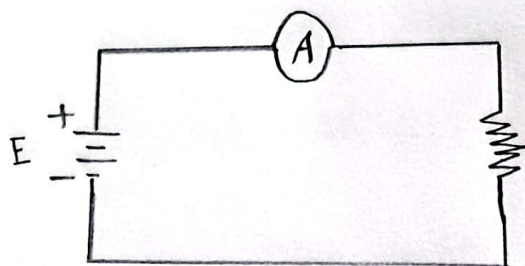


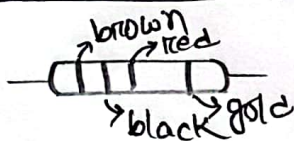
Figure: Circuit diagram to verify Ohm's law

③

1. Connect all the components on the bread board according to the figure.
2. Turn the DC power supply on and set the voltage to 5V
3. Measure the current and resistance using multimeter and note them down.
4. Repeat last two steps for the voltage of 10V and 15V.
5. The process is same for remaining two resistors.
6. Compare the measured value with calculated value.

Result and Discussion:

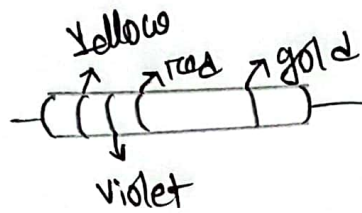
Resistor 1:



measured
Value(R_m) = $10 \times 10^2 \pm 5\%$
= $1000 \Omega = 1k\Omega$

$V(V)$	$I(mA)$	$R(k\Omega)$	R_m
5	5.3	0.943	1k Ω
10	10.7	0.9345	
15	16.1	0.931	

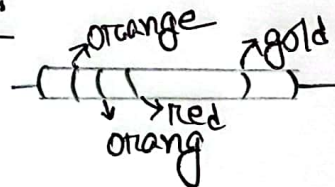
Resistor 2:



$$\begin{aligned}\text{measured value } (R_m) &= 47 \times 10^2 \pm 5\% \\ &= 4700 \Omega \\ &= 4.7 \text{ k}\Omega\end{aligned}$$

V(v)	I(mA)	R(k Ω)	R _m (k Ω)
5	1.1	4.5	4.7 k Ω
10	2.3	4.35	
15	3.4	4.412	

Resistor 3:



$$\begin{aligned}\text{measured value } (R_m) &= 33 \times 10^2 \pm 5\% \\ &= 3300 \Omega = 3.3 \text{ k}\Omega\end{aligned}$$

V(v)	I(mA)	R(k Ω)	R _m
5	1.6	3.13	3.3 k Ω
10	3.2	3.125	
15	4.8	3.125	

Conclusion: The calculated resistance may have error. This mostly due to the problem of apparatus. Also may be due to the loose connection. Although voltage and current was measured using digital multimeter, reading was fluctuating and it was difficult to take exact reading.